

Stage 3: Head-Level Characterization

Revised experimental plan

Phase A: Single-hop SIH Audit

21 September 2026

1 Stage 3: Head-Level Characterization

1.1 Question, scope and outcome

Stage 3 asks why RI-based selection and single-head intervention effects disagree, and what the selected heads actually do in the single-hop task. Stage 2 found several large effects outside the RI selection, small effects for several RI priorities, and definition-dependent rank correlations. These observations do not establish copying, semantic specificity, or a writer–reader circuit.

For each head we seek an evidence-based profile: *where its activity affects the answer; what it attends to; what its contextual output writes; and which alternative explanations remain*. A familiar type, a precisely described new pattern, or an unexplained result are all acceptable outcomes. We do not require every head to receive a categorical label.

All work uses discovery families. The 87 validation families and the location world remain untouched. Inter-head paths, joint head interventions, circuits and faithfulness belong to Stage 4. The layer-wise residual divergence analysis is also moved to Stage 4 as a descriptive guide, not a constraint that rules out important computation in later layers. The proposed weight-only relation classifier is removed: fictitious child–mother assignments are supplied by the context, not fixed associations in the weights.

1.2 Fixed head inventory

Let I_h be the saved exact Scope-P mean on the same 178 pairs (89 families, orders averaged), with positive sign denoting movement toward the corrupted answer. Use only the 148 heads with this coverage; never substitute a 40-pair mean for a missing 178-pair mean. Absence from this population does not imply a small effect.

Let R be the union of the 59 test-only RI candidates and the historical pair L3H11/L9H22. This denotes selection by the recorded rules, not a calibrated SIH discovery or a universally high RI score. The historical top-25 extension rankings are not added to R .

The thresholds below are descriptive choices made after inspecting Stages 1–2 and before Stage-3 measurements. They are not significance thresholds or evidence of irrelevance. Membership depends on the written rule, with no named exceptions.

RI-selected, small mean effect (48 heads).

$h \in R$ and $|I_h| < 0.1$. Includes the historical pair and L11H4, L17H5 and L9H16. A small signed mean can conceal heterogeneous effects.

Strong positive effect outside RI selection (12 heads).

$h \notin R$ and $I_h \geq 0.3$. Includes L17H1, L27H6 and L18H18. “Outside selection” is not synonymous with a supported low RI score.

Strong positive effect within RI selection (5 heads).

$h \in R$ and $I_h \geq 0.3$: L16H1, L16H21, L17H24, L18H19 and L22H5. Final-position QK agreement is recorded separately, not assumed for this group.

Strong negative effect (8 heads).

$I_h \leq -0.3$, irrespective of RI membership. Record RI membership separately. These are negative intervention effects, not established negative name movers.

Random reference heads (25 heads).

The unchanged Stage-2 random controls. Preserve their original selection provenance; they were sampled outside the original deterministic shortlist. They are descriptive references, not a calibrated null population.

Moderate-effect supplement (7 heads). Include all heads in the same 148-head population with $0.1 \leq |I_h| < 0.3$: L6H24, L7H1, L8H15, L12H17, L14H23, L15H3 and L16H31. This explicit band retains informative intermediate effects without relabeling them as strong. L9H16 remains in the small-effect RI group under the same rule.

The inventory contains **105 unique heads**, including all 61 members of R and all 25 controls. A head receives every applicable label; no head is added by example alone. There are no overlaps between the five groups and the supplement in the current snapshot. RI membership and final-position QK support remain separate attributes. The audit files are `results/stage3_plan/head_inventory.csv` (all 148 heads, including exclusions) and `results/stage3_plan/inventory_policy.json` (rules, lists and source hashes). Only inventory calculations have been performed; no Stage-3 effects were computed.

1.3 Required analyses, in order

The order follows dependencies: establish what is known, locate the effect, then interpret reading and writing at the relevant sites. It is not a requirement to load the model separately for each analysis. Compatible captures should share forward passes and be saved for reuse.

Evidence Review and Logit-Change Decomposition

Purpose. Establish the measured effects and alternative explanations already available, without treating a previous narrative report as primary data.

Procedure. Assemble saved per-head Scope-P/Scope-F effects, family and order variation, RI selection reasons, and event support. Distinguish unavailable Scope-F results from zero effects. For available Scope-F readouts report

$$P_h = z_{\text{corr}}^{\text{patched}} - z_{\text{corr}}^{\text{clean}}, \quad S_h = z_{\text{clean}}^{\text{clean}} - z_{\text{clean}}^{\text{patched}}, \quad I_h^F = P_h + S_h.$$

These describe the logit changes caused by replacing one activation with another; they do not identify a head's direct promotion or suppression mechanism. The existing child-name readout uses its first token. Flag shared-prefix cases and token collisions; do not interpret it as whole-name probability or as a validated specificity measure.

Validity prerequisite. Resolve the previously identified cross-sample `residual_40` and family-bootstrap indexing errors before using those statistics. Reuse valid measurements; do not silently recompute missing or invalid analyses. Record the correction task separately from the experiment. Missing attention or residual caches must be identified before classifying work as CPU-only.

Causal Position Profile and Reverse Patching

Purpose. Locate the sites where a head’s activity influences the answer, without assuming that it is a fact-position writer or an answer-position reader.

Procedure. Preserve the Stage-2 outcome: the first divergent answer-token logit contrast with a fixed clean-answer sign and any shared answer prefix. Complete missing Scope-F measurements for all inventory heads on the 178 pairs; 46 inventory heads currently lack them. Scope F always patches the last original prompt token, not an appended answer-prefix position. Save the two answer logits and their clean baselines for every new intervention.

On the original common 40 pairs, patch each head separately at each test-block position from the first differing token through the final colon. Earlier identical prefix positions are checked with neutral probe patches rather than exhaustively recomputed. Annotate positions by fact, entity role, token anchor, question and punctuation. Report the position profile, both signed means and family variation, with Scope P and F compared on these same 40 pairs. Do not subtract P and F to claim an additive contribution of earlier positions.

For the ten largest $|I_h|$ inventory heads, fixed from the saved 178-pair means, also perform reverse patches (clean donor into corrupted recipient) at all prompt positions and at the final colon on all 178 pairs. Report $M(x_r; a_h \leftarrow a_h^c) - M(x_r)$ alongside the noising effect. Recovery here is conditional on the rest of the corrupted model; it is not proof that the head alone suffices to solve the task. The fixed ten-head list is in the inventory policy.

The position scan is the expensive component. Before submission, measure runtime on probe pairs and report the total work for this fixed scope. Any later reduction or expansion of scope must be documented, not introduced by dropping costly heads.

Relation-Tracking Attention Profile

Purpose. Test whether attention selects relevant entities or follows positions, token identities, punctuation, or local offsets.

Procedure. Inspect the final colon and test-block rows, marking the causally sensitive sites from the position scan. Use the 20 common discovery families with both orders and base/corrupted/reorder variants. Record attention to the child and mother separately, both over each full name span and at the Stage-1 anchors; compare with the same roles in distractor facts. Record argmax, relative offset, self/previous-token attention and the original RI dominance gate. Analyze all eligible rows, including QK failures. Saved passing events alone cannot supply this unconditional profile; capture missing attention in new runs.

For each base prompt in these families, add a paired query asking about the other answer-side child while leaving facts unchanged. Re-annotate the relevant fact and report the model’s response on this new condition; do not select cases by whether the desired attention pattern appears. This query control is a new discovery measurement, distinct from the original corruption.

Use reordering to test sensitivity to physical location and the query control to test relevance to the question. In the original corruption, mothers change at fixed fact positions: correct tracking can preserve the same attention location. Report paired role-specific attention differences and support; no positional or semantic label follows from argmax movement alone. Tokenization-length changes are handled through spans and roles, not assumed position alignment.

Contextual Output Profile

Purpose. Describe what each head writes at the positions examined above, and compare this with the raw-embedding diagnostic used by RI.

Procedure. On the same clean and corrupted inputs, record the actual per-head output-projection contribution o_j^h , using the model’s real value inputs and attention weights. This is the head’s contribution before any subsequent attention-output normalization; such normalization is shared across heads and must not be treated as an additive per-head operation. Inspect $o_j^h W_U$ for the current token and the annotated first/last tokens of the target, source and other visible names. Use the same token IDs for raw/contextual comparisons, retain collision ties, and report ranks and signed contrasts rather than “logit shares.”

This vocabulary projection is a diagnostic, not the final model distribution or the causal effect. Do not apply final normalization to an isolated head vector and call the result an exact logit contribution. A later normalization-aware variant must specify its fixed reference or local linearization explicitly. Keep the bare-colon site separate from the later metric site in shared-prefix cases. An output at an early site need not encode the final answer directly.

1.4 Planned diagnostic analyses after the core profiles

All four analyses below are included in Stage 3; they are not alternatives selected only after a promising result. Apply them to the fixed inventory, including the random reference heads, wherever their inputs and sites are well defined. Record the head list, input set, sites, controls and scoring protocol before each analysis. Missing support is reported as insufficient data, never as a zero score or a negative finding. Eligibility must not depend on obtaining the expected result.

The core profiles establish the relevant sites and interpretation. Weight-based copying and synthetic fingerprints then test complementary static and behavioral properties; contextual RI tests the embedding choice; attention–value interventions test routing and content causally. These analyses may share captures and run in parallel where independent; their scientific scope is not reduced by this scheduling.

Copying and Token-Preference Diagnostic. **Purpose and scope:** test copying and lexical preference in every inventory head, and whether current-name/control overlap helps explain RI contrasts. First partition the available saved events by whether the current token belongs to the target, a control name, or neither; report support and target/control scores in each partition. Then inspect the weight map $e_t W_V^h W_O^h W_U$: self-token rank and self-versus-other-name contrast, with equal weight per distinct token ID. Include a fixed non-name reference sample to distinguish name-token behavior from broader copying. The weight diagnostic does not require QK passes; report missing event-based comparisons separately. Positive or negative contrast is descriptive, not an established mover type. Broad spectral analysis is not required; weight alignment alone cannot decide the copy-penalizing hypothesis.

Attended-name mover classification. The weight map describes the current token; the complementary question is what the head does to the token it attends to. Using the saved test-block records of the common families (attention argmax and the contextual output projection $o_j^h W_U$ over the visible name tokens at the same position), select for each head the events whose argmax falls on a name token and compute the contextual logit of that attended name relative to the mean of the other visible names. A head is labeled a *name mover* if this contrast is positive in at least 80% of its events, and a *negative name mover* if it is negative in at least 80%, subject to the fixed support rule (≥ 20 events, ≥ 10 families); heads below support are reported as insufficient, not as neither. The label must be cross-checked against the Scope-F logit decomposition: a negative name mover is expected to lower the clean-answer logit when patched, so a mismatch between the two signs is reported as a finding rather than resolved by relabeling. This classification uses only records already produced by the core capture and requires no additional measurement. The candidates *L23H15*, *L19H16* and *L26H23* are checked first but the classification is applied to the full inventory, including the random controls.

Matched Contextual RI Comparison. **Purpose and scope:** compare raw and contextual RI for every inventory head with valid saved QK events, whether or not its core profiles suggest disagreement. Retain the extension’s metric-specific support rule (at least 20 comparisons across 10 families) for summary rankings and correlations; retain smaller samples as explicitly insufficient-support descriptives. Replace only the raw embedding in the RI projection by the contextual activation entering the head’s value projection, including the model’s actual pre-value normalization. Hold event IDs, visible-token normalization, controls, anchors and family aggregation fixed. Undefined scores are reported and matched comparisons use common valid support. This current-position projection is still distinct from the attention-weighted head output. Compare raw and contextual scores on the same heads and pairs, reporting eligibility. It cannot repair missing QK events; a correlation in this selected inventory is not an all-1,024-head result. This is a sensitivity analysis, not a claimed novel method.

Attention–Value Intervention. **Purpose and scope:** test routing, values and their interaction for every inventory head at the final colon, including heads with small effects. Additional sites may be selected from the causal position profile under a documented rule fixed before these interventions; label those results as post-selection exploration. At each specified site compare clean attention/clean values, corrupted attention/clean values, clean attention/corrupted values, and corrupted attention/corrupted values. Use captured post-softmax attention and actual value activations while preserving the model’s other computations. Gate the joint replacement against the ordinary head-output patch at that site. Report both single replacements and the interaction between them; do not force an additive or pattern/value-dominated classification.

Repeated-Sequence and Retrieval Fingerprints. **Purpose and scope:** test all inventory heads on both a repeated-sequence benchmark (prefix matching and copying) and a retrieval benchmark (attention to the copied token during defined copy events). Neither benchmark is conditional on a prior type label. Fix their token sampling, lengths, seeds, event definitions, handling of repeated token occurrences and scoring protocol before measurement. Run both attention and output checks with the same reference heads. Report task success and event denominators, including heads with insufficient eligible events. Transfer to a synthetic task supports a behavioral capability, not proof that the same algorithm solves the kinship task. Function-vector mediation and other additional fingerprints are not required; they need a separately justified task-level question and protocol.

1.5 Optional extensions

The following readout remains conditional. Broad spectral copying analysis, function-vector mediation and additional synthetic benchmarks may also be proposed separately when they answer a remaining question; none replaces the four planned analyses above or is required for completion.

Controlled Representation Readout (Patchscopes-style). **Trigger:** a causal site is localized but vocabulary projections cannot explain the information present there. This remains optional within Stage 3, not automatically assigned to Stage 4. A residual-stream readout describes a multi-component representation; it cannot be attributed to one head merely because it was captured near that head. Compare intact and head-intervened representations in the same readout prompt, with no-injection, irrelevant-injection and entity-swap controls. Treat decoded text as supporting exploration, not a causal verdict; fictitious names do not rule out lexical copying or information from the readout prompt. Questions about transfer between heads belong to Stage 4.

1.6 Reporting, completion and handoff

Average orders within family and report family support, heterogeneity and missing data. Random controls provide context; their 95th percentile is not a universal property-detection or significance threshold. Any uncertainty interval must name its population, resampling unit, number of draws and quantities held fixed. Do not reuse an invalid interval or turn exploratory selection into confirmation.

Deliver a head-profile table and short evidence cards covering inventory reasons, position effects, attention roles, contextual output, alternative explanations, and results of all four planned diagnostic analyses. For each head and analysis, distinguish completed, insufficient support, unavailable input and unresolved implementation; unsupported cells must not be silently omitted or zero-filled. Track optional extensions separately (not triggered, completed, or unresolved). Completion requires the core profiles and all four planned analyses on their applicable populations, with exclusions documented; unavailable inputs and implementation failures remain outstanding rather than counting as completion. It does not require favorable results or a type label for every head.

The copying, contextual-information and early/late-position hypotheses receive separate evidence assessments, not forced yes/no verdicts. Stage 4 receives candidate components and specific communication hypotheses to test with path and group interventions, then faithfulness and held-out validation. Neither a similar fingerprint nor a large single-head effect establishes an edge or circuit.